Practice workbook · Finding r from sums of a geometric series

Writing given sums (and terms) of a geometric series in a and r, eliminating a, and solving the resulting quadratic or cubic in r.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-10

1. Question 8 · part a · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

8 The sum of the first and second terms of a geometric series G is 400

The sum of the second and third terms of G is 100

(a) Show that the common ratio of G is $\frac{1}{4}$

(4)

(b) Show that the first term of G is 320

(2)

(c) Find the sum to infinity of G

(2)

The sum to n terms of G is S_n

(d) Find, using logarithms, the least value of n such that

$$S_n > 426.6$$

(4)

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1. Question 8 · part a · 4 marks · Working and answer

Working and answer

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2. Question 5 · part a · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

5 The nth term of a geometric series with common ratio r is u_n

Given that $u_2 + u_4 = 212.5$ and that $u_3 + u_4 = 62.5$

(a) find the two possible values of r.

(5)

Given that the series is convergent with sum to infinity S,

(b) find the exact value of S.

(2)

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2. Question 5 · part a · 5 marks · Working and answer

Working and answer

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3. Question 7 · part b · 5 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

7 The 7th term of a geometric series is 192 and the 8th term of this geometric series is 1152

(a) Find, as a fraction in its simplest form, the 4th term of this geometric series.

(3)

A different geometric series G has a common ratio r and nth term t_n

Given that $t_3 = 24$ and $t_2 + t_3 + t_4 = -36$

(b) show that r satisfies the equation

$$2r^2 + 5r + 2 = 0$$

(5)

Given further that G is convergent with sum to infinity S,

(c) find the value of S.

(4)

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3. Question 7 · part b · 5 marks · Working and answer

Working and answer

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4. Question 8 · part b · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part b only.

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The sum of the second and third terms of G is 100

(a) Show that the common ratio of G is $\frac{1}{4}$

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(b) Show that the first term of G is 320

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The sum to n terms of G is S_n

(d) Find, using logarithms, the least value of n such that

$$S_n > 426.6$$

(4)

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4. Question 8 · part b · 2 marks · Working and answer

Working and answer

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5. Question 8 · part a · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

8 The sum of the first 2 terms of a geometric series G is 360
The sum of the 2nd and 3rd terms of G is 288

The nth term of G is U_n

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where A is an integer to be found.

(7)

(b) Explain why G is convergent.

(1)

(c) Hence find the sum to infinity of G

(2)

(d) Find the least number of terms for which the sum is greater than 978

(4)

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5. Question 8 · part a · 7 marks · Working and answer

Working and answer

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6. Question 9 · part a · 8 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

9 A geometric series G has common ratio r where $r > 0$

The third term of G is $\frac{27}{2}$ and the sum of the first three terms of G is $\frac{57}{2}$

Given that the sum to n terms of G is S_n

(a) show that $S_n = \sum_{j=1}^{n} 4\left(\frac{3}{2}\right)^j$

(8)

Given that $S_k > 50\,000$

(b) show that the least value of k is given by

$$k > \frac{\lg\left(\frac{12\,503}{3}\right)}{\lg\left(\frac{3}{2}\right)}$$

(3)

(c) Hence find the least value of k

(1)

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6. Question 9 · part a · 8 marks · Working and answer

Working and answer

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7. Question 7 · part a.i · 0 marks

2021 NOV · Paper 02

Further practice · Attempt part a.i only.

7 A geometric series G has first term a and common ratio r

The sum of the first three terms of G is $\frac{61}{6}$

The sum to infinity of G is $\frac{125}{6}$

(a) (i) Show that $r = \frac{4}{5}$

(ii) Find the value of a

(6)

The sum of the first n terms of G is S_n

Given that $S_n > 19.8$

(b) show that $n \lg\left(\frac{4}{5}\right) < \lg\left(\frac{31}{625}\right)$

(2)

(c) Hence find the least value of n

(2)

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7. Question 7 · part a.i · 0 marks · Working and answer

Working and answer

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8. Question 7 · part a.ii · 6 marks

2021 NOV · Paper 02

Further practice · Attempt part a.ii only.

7 A geometric series G has first term a and common ratio r

The sum of the first three terms of G is $\frac{61}{6}$

The sum to infinity of G is $\frac{125}{6}$

(a) (i) Show that $r = \frac{4}{5}$

(ii) Find the value of a

(6)

The sum of the first n terms of G is S_n

Given that $S_n > 19.8$

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(2)

(c) Hence find the least value of n

(2)

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8. Question 7 · part a.ii · 6 marks · Working and answer

Working and answer

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9. Question 7 · part c · 3 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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9. Question 7 · part c · 3 marks · Working and answer

Working and answer

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10. Question 7 · part d · 1 mark

2024 NOV · Paper 02

Further practice · Attempt part d only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
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(4)

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10. Question 7 · part d · 1 mark · Working and answer

Working and answer

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11. Question 7 · part e · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part e only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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11. Question 7 · part e · 2 marks · Working and answer

Working and answer

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12. Question 4 · part whole · 7 marks

2022 JUN · Paper 01

Further practice · Attempt part whole only.

4 The common ratio of a geometric series G is positive.

The sum of the first 4 terms of G is 80
The sum to infinity of G is 81

Show that the sum of the first 7 terms of G differs from the sum to infinity of G by $\frac{1}{27}$

(7)

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12. Question 4 · part whole · 7 marks

2022 JUN · Paper 01

Further practice · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 7 marks)

12. Question 4 · part whole · 7 marks · Working and answer

Working and answer

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13. Question 7 · part a · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

7 A geometric series G with common ratio r, has first term 16 and third term $\frac{2704}{625}$

(a) Find the two possible values of r

(2)

Given that $r > 0$

(b) find the sum to infinity of G

(2)

The sum to n terms of G is greater than 33

(c) Find, using logarithms, the least possible value of n
Show your working clearly.

(5)

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13. Question 7 · part a · 2 marks · Working and answer

Working and answer

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